

FINAL PROJECT DUE JUNE 18, 2026 by 4PM

1 Definition 1: TLS

We define traces that establish a TLS handshake between a client C and a server S . Let the following be terms within the trace:

$$\begin{aligned} C, S, CA &\in Name \\ r_C, r_S &\in Nonce \\ e_S, e_C &\in Expt \end{aligned}$$

$$\begin{aligned} Cert_S &= S \parallel pk(S) \parallel enc(hash(S, g^{e_S}), sk(CA)) \\ S_{KE} &= g^{e_S} \parallel enc(hash(r_C \parallel r_S \parallel g^{e_S}), sk(S)) \\ PMS &= g^{e_S \times e_C} \\ C_{WRITE} &= hash(PMS, r_C, r_S, \text{"client-write"}) \\ S_{WRITE} &= hash(PMS, r_C, r_S, \text{"server-write"}) \\ C_{FMESG} &= r_C \parallel r_S \parallel Cert_S \parallel S_{KE} \\ C_{FIN} &= enc(hash(PMS \parallel \text{"client-fin"} \parallel C_{FMESG}), C_{WRITE}) \\ S_{FIN} &= enc(hash(PMS \parallel \text{"server-fin"} \parallel C_{FMESG} \parallel C_{FIN}), S_{WRITE}) \end{aligned}$$

Let trace $Tr_{TLS-C}(C, S, CA, r_C, r_S, e_S, e_C)$:

1. $+ r_C$
2. $- r_S \parallel Cert_S \parallel S_{KE}$
3. $+ g^{e_C} \parallel C_{FIN}$
4. $- S_{FIN}$

Let the complementary trace $Tr_{TLS-S}(C, S, CA, r_C, r_S, e_S, e_C)$:

1. $- r_C$
2. $+ r_S \parallel Cert_S \parallel S_{KE}$
3. $- g^{e_C} \parallel C_{FIN}$
4. $+ S_{FIN}$

2 Definition 2: SCEP Token

Let

$$Tok_{X,Y} = \text{“Token”} \parallel X \parallel pk(X) \parallel Y \parallel pk(Y) \parallel data$$

be the actual token. We then define the actual SCEP token as:

$$T_{X,Y} = Tok_{X,Y} \parallel enc(hash(Tok_{X,Y}), sk(Y))$$

3 Definition 3: SCEP

We define traces that carry out SCEP between $\mathcal{S}$ and a responding infrastructure server K or H , which we define as $\mathcal{W}$. As before, we first define the terms that we use within the traces:

$$\begin{aligned} \mathcal{S}, \mathcal{W}, M, CA &\in Name \\ r_{\mathcal{S}}, r_{\mathcal{W}}, r'_{\mathcal{S}}, r'_{\mathcal{W}}, \omega &\in Nonce \\ e_{\mathcal{S}}, e_{\mathcal{W}} &\in Expt \end{aligned}$$

$T_{\mathcal{S},M}$ and $T_{\mathcal{W},CA}$ are as in Definition 2.

$$\begin{aligned} \mathcal{S}_{WRITE} &= hash(g^{e_{\mathcal{S}} \times e_{\mathcal{W}}}, r'_{\mathcal{S}}, r'_{\mathcal{W}}, \text{“client-write”}) \\ \mathcal{W}_{WRITE} &= hash(g^{e_{\mathcal{S}} \times e_{\mathcal{W}}}, r'_{\mathcal{S}}, r'_{\mathcal{W}}, \text{“server-write”}) \end{aligned}$$

Let $Tr_{SCEP-SW}(\mathcal{S}, \mathcal{W}, M, CA, r_{\mathcal{S}}, r_{\mathcal{W}}, r'_{\mathcal{S}}, r'_{\mathcal{W}}, \omega, e_{\mathcal{S}}, e_{\mathcal{W}})$:

1. $Tr_{TLS-C}(\mathcal{S}, \mathcal{W}, CA, r'_{\mathcal{S}}, r'_{\mathcal{W}}, e_{\mathcal{S}}, e_{\mathcal{W}})$ (Definition 1)
2. $+ enc(r_{\mathcal{S}} \parallel T_{\mathcal{S},M}, \mathcal{S}_{WRITE})$
3. $- enc(\omega \parallel r_{\mathcal{W}} \parallel T_{\mathcal{W},CA} \parallel enc(hash(\text{“server-mutauth”, } r_{\mathcal{S}}, r_{\mathcal{W}}, T_{\mathcal{W},CA}), pk(\mathcal{W})), \mathcal{W}_{WRITE})$
4. $+ enc(\omega \parallel enc(hash(\text{“client-mutauth”, } r_{\mathcal{S}}, r_{\mathcal{W}}, T_{\mathcal{S},M}), pk(\mathcal{S})), \mathcal{S}_{WRITE})$

Let $Tr_{SCEP-WS}$ be complementary to $Tr_{SCEP-SW}$, such that $Tr_{SCEP-WS}$ (1) includes Tr_{TLS-S} rather than Tr_{TLS-C} , and (2) inverts the directions of terms 2, 3, and 4.